The Geometry of Spacetime (GR Club)

§2.4 Physical Consequences, Ling-Wei Luo.

Length and Time.

Length and time are no longer indep.

For that observer, all events on the worldline differ only in the time they happen, not the place.

$$1 = c = 2.9979246 \times 10^8 \frac{m}{s}$$

$$\Rightarrow 1s = 2.9979246 \times 10^8 m$$

$$\Rightarrow \boxed{1m = 3.3356409 \cdot 10^{-9} s}$$

geometric unit

$$1 = G = 6.67 \times 10^{-11} \frac{m^3}{kg \cdot s^2}$$

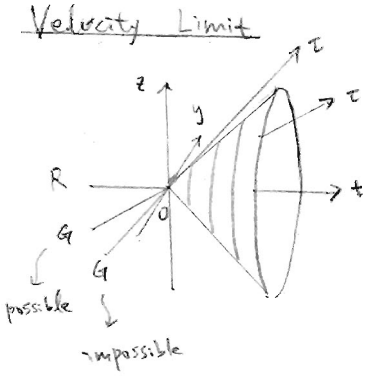
$$\Rightarrow 1\,kg = 6.67 \times 10^{-11} \frac{m^3}{s^2} = 2.48 \times 10^{-36} s$$

$$\Rightarrow [\text{velocity}] = 0$$

$$\Rightarrow [\text{energy}] = [\text{momentum}] = [M] = [T]$$

$$\Rightarrow [\text{acceleration}] = [T^{-1}]$$

Velocity Limit



G's worldline must be lie inside the light cone.

$\Rightarrow v < c$

Simultaneity (Not Physical Meaningful)

Proposition 2.12

Given any spacelike event E whatsoever, there is an observer who says that E and O are simultaneous

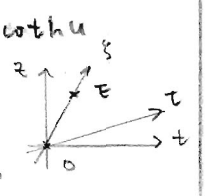
pf Spacelike: $t^2 - z^2 = -k^2 < 0 \Rightarrow z^2 - t^2 = k^2$

$\Rightarrow (t, z) = (k\sinh u, k\cosh u)$

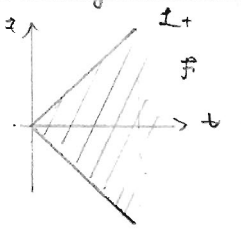
Let G have velocity $v = \frac{k\cosh u}{k\sinh u} = \coth u$ relative to R.

$\Rightarrow$ E will lie on G's ξ-axis

$\Rightarrow$ E and O both happen at time $\tau = 0$



The Objective Future



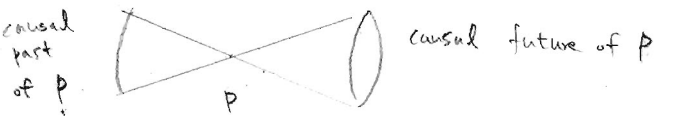
Objective future: the common future of all observers

Objective past: the common past of all observers

Causality.

Def 2.10

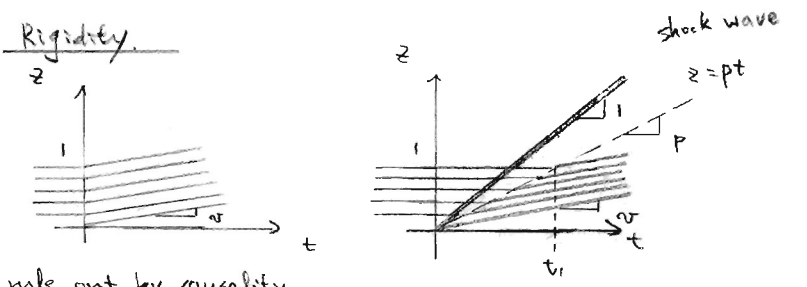
The causal future of an event P is the set of all events that P can influence. The causal past of P is the set of all events that can influence P.



all the events that P can influence lie on worldlines that have slope v where $|v| \le 1$.

$\Rightarrow$ Causality cannot travel faster than the speed of light.

Rigidity.



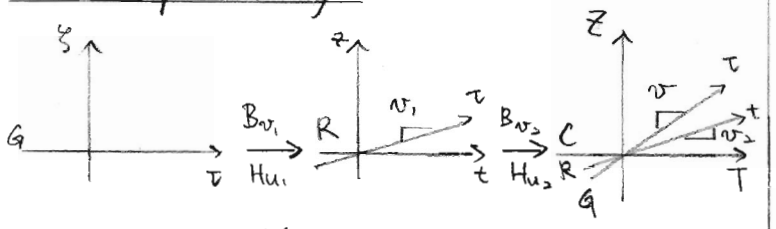
rule out by causality.

The simplest assumption $t \propto z \Rightarrow z = pt$.

$|v| < |p| < 1$

. The entire rod is motion after $t = t_1$

Addition of Velocity



$v = -2/3$ $\qquad$ $v = 2/3$

galaxy C $\qquad$ earth R $\qquad$ galaxy G

$B_{v_1} : G \dashrightarrow R$

$B_{v_2} : R \dashrightarrow C$ $\quad \Rightarrow \quad B_v = B_{v_2} B_{v_1} : G \text{ —— } C$

convert the boosts to hypebolic rotations

$\Rightarrow H_u = B_v$ where $v = \tanh u$

If $H_u = H_{u_2} H_{u_1} \longrightarrow u = u_1 + u_2$

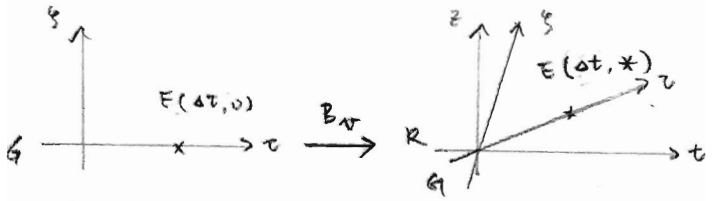
$\therefore v = \tanh u = \tanh(u_1 + u_2)$

$$= \frac{\tanh u_1 + \tanh u_2}{1 + \tanh u_1 \tanh u_2} = \frac{v_1 + v_2}{1 + v_1 v_2}$$

p.44 addition formulas

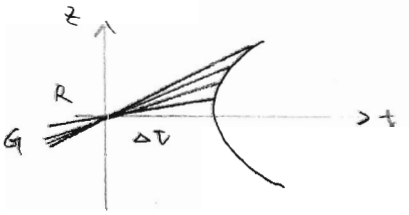
$|v| = |\tanh u| < 1.$

Moving Clocks Run Slow (Time Dilation)

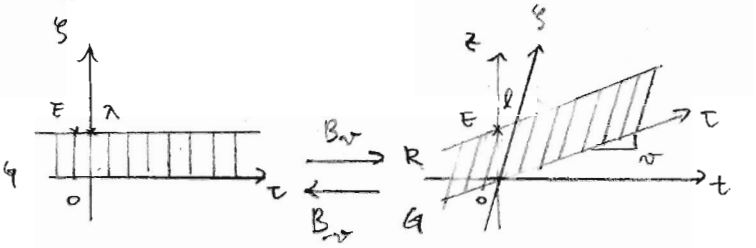


$$\begin{pmatrix} \Delta t \\ * \end{pmatrix} = B_v \begin{pmatrix} \Delta\tau \\ 0 \end{pmatrix} = \frac{1}{\sqrt{1-v^2}} \begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix} \begin{pmatrix} \Delta\tau \\ 0 \end{pmatrix} = \begin{pmatrix} \Delta\tau / \sqrt{1-v^2} \\ * \end{pmatrix}$$

time dilation : the calibration of different time axes by hyperbolic rotations.



Moving Rulers Contract in Length (Lorentz Contraction)



$$\begin{pmatrix} * \\ \lambda \end{pmatrix} = B_{-v} \begin{pmatrix} 0 \\ l \end{pmatrix} = \frac{1}{\sqrt{1-v^2}} \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \begin{pmatrix} 0 \\ l \end{pmatrix} = \begin{pmatrix} * \\ l / \sqrt{1-v^2} \end{pmatrix}$$

$\Rightarrow l = \sqrt{1-v^2}\, \lambda$

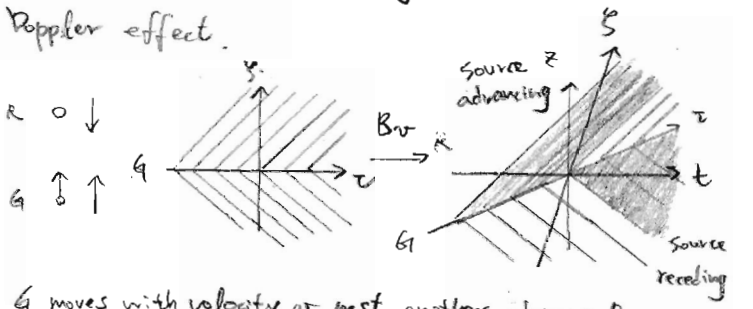
$\Rightarrow$ Fitzgerald contraction

(or Lorentz contraction)

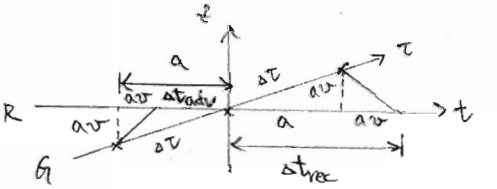
The Doppler Effect

The frequency of a light signal depend on whether the source is moving.

$\Rightarrow$ Doppler effect.



As G moves with velocity v past another observer R, the source is first advancing toward R, the receding.



in G $\Delta\tau = 1/\nu \Rightarrow \nu = 1/\Delta\tau$

in R frequency $n = 1/\Delta t$

$\because \Delta\tau^2 = a^2 - a^2 v^2 \Rightarrow a = \Delta\tau / \sqrt{1-v^2}$

advancing side:

$$\Delta t_{adv} = a - av = \Delta\tau \frac{1-v}{\sqrt{1-v^2}} \Rightarrow n_{adv} = \frac{1}{\Delta t_{adv}} = \nu \sqrt{\frac{1+v}{1-v}} > \nu$$

receding side:

$$\Delta t_{rec} = a + av = \Delta\tau \frac{1+v}{\sqrt{1-v^2}} \Rightarrow n_{rec} = \frac{1}{\Delta t_{rec}} = \nu \sqrt{\frac{1-v}{1+v}} < \nu$$

$\Rightarrow v \begin{cases} > 0, & \text{source and observer are approaching} \\ < 0, & \text{source and observer are separating} \end{cases}$

$$\Rightarrow n = \nu \sqrt{\frac{1-v}{1+v}} \xrightarrow[|v| \ll 1.]{\text{classical limit}} n \approx \nu(1-v)$$

General Relativity Study Club
in Summer Semester 2009

Jason Ling-Wei Luo
Department of Physics, National Tsing Hua University

1 Physical Consequences

1.1 Length and Time

Length and time are no longer independent. For that observer, all events on the worldline differ only in the *time* they happen, not *place*.

1.2 Velocity Limit

G's worldline must lie *inside* the light cone $\Rightarrow |v| < 1$.

1.3 Simultaneity

Definition: *Given any spacelike event E whatsoever, there is an observer who says that E and O are simultaneity.*

The principle of relativity implies that the notion of simultaneity is not physically meaningful.

1.4 The Objective Future

We call the common future of all observers the *objective future* of O.

1.5 Causality

Definition: *The **causal future** of an event is the set of all events that A can influence. The **causal past** of A is the set of all events that A can influence.*

1.6 Rigidity

In the scenario, the far end of the rigid object can not move immediately, it rules out by the principle of causality.

1.7 Addition of velocities

Hyperbolic rotation:
The relation is $H_u = B_v$, where $v = \tanh u$ and $u = \tanh^{-1} v$. If $H_u = H_{u_1} H_{u_2}$

then $u = u_1 + u_2$, and thus

$$v = \tanh u = \tanh(u_1 + u_2) = \frac{\tanh u_1 + \tanh u_2}{1 + \tanh u_1 \tanh u_2} = \frac{v_1 + v_2}{1 + v_1 v_2} < 1,$$
$$\Rightarrow |v| = |\tanh u| < 1.$$

1.8 Moving Clocks Run Slow

$$\begin{pmatrix} \triangle t \\ * \end{pmatrix} = B_v \begin{pmatrix} \triangle \tau \\ 0 \end{pmatrix} = \frac{1}{\sqrt{1-v^2}} \begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix} \begin{pmatrix} \triangle \tau \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{\triangle \tau}{\sqrt{1-v^2}} \\ * \end{pmatrix}.$$

Thus $\triangle t = \triangle \tau / \sqrt{1-v^2} > \triangle \tau$.
$\Rightarrow$ Time dilation.

1.9 Moving Rulers Contract in Length

$$\begin{pmatrix} * \\ \lambda \end{pmatrix} = B_{-v} \begin{pmatrix} 0 \\ l \end{pmatrix} = \frac{1}{\sqrt{1-v^2}} \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \begin{pmatrix} 0 \\ l \end{pmatrix} = \begin{pmatrix} * \\ \frac{l}{\sqrt{1-v^2}} \end{pmatrix}.$$

Then we obtain $l = \sqrt{1-v^2}\lambda$.
$\Rightarrow$ Length contraction.

1.10 The Doppler Effect

Although the light signal does not depend on whether the source is moving, its *frequency* does. This is the Doppler effect. The frequency to the observer R is

$$n = \nu \sqrt{\frac{1-v}{1+v}}.$$

Classical limit ($|v| \ll 1$):

$$\sqrt{\frac{1-v}{1+v}} \approx 1 - v.$$